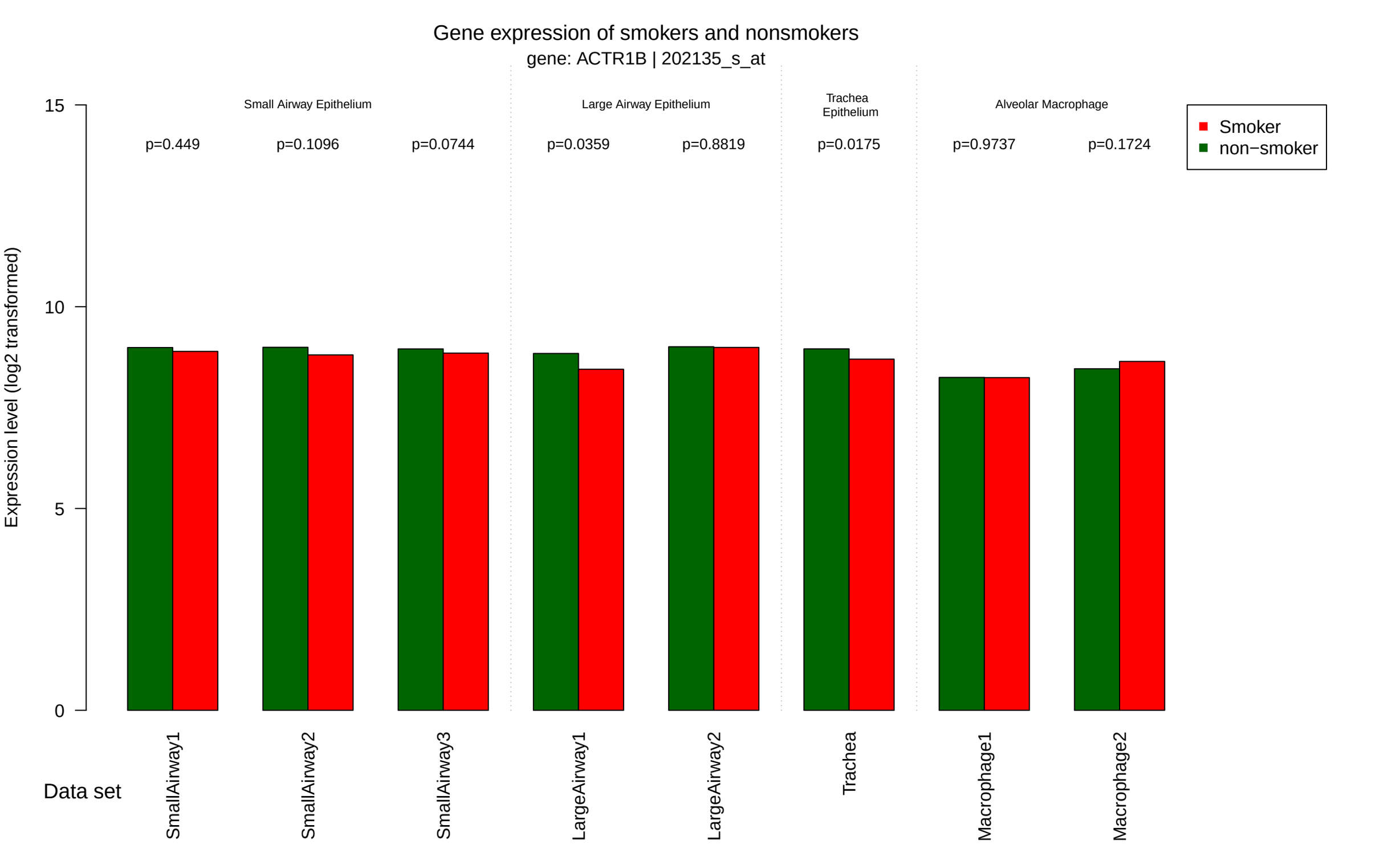
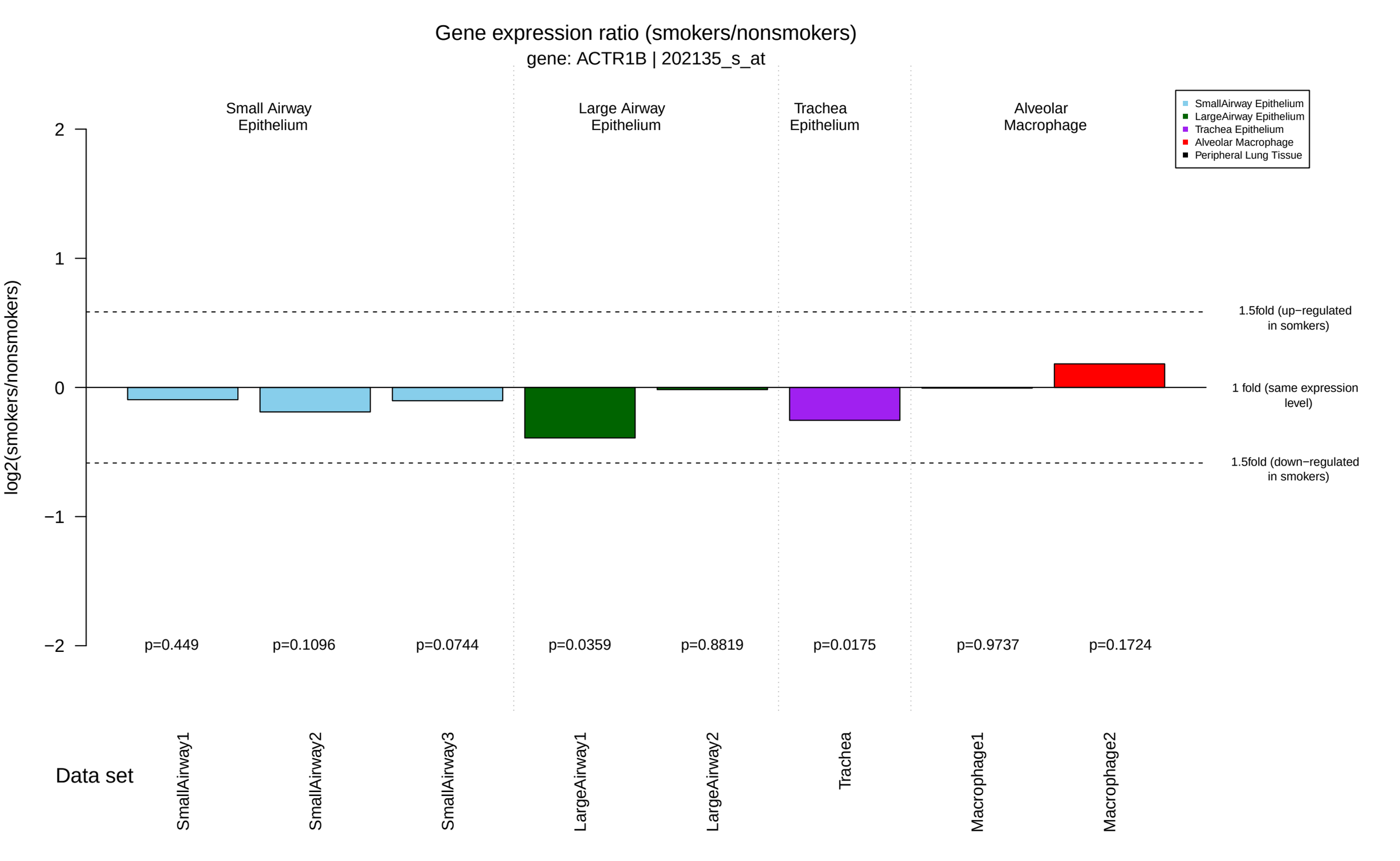
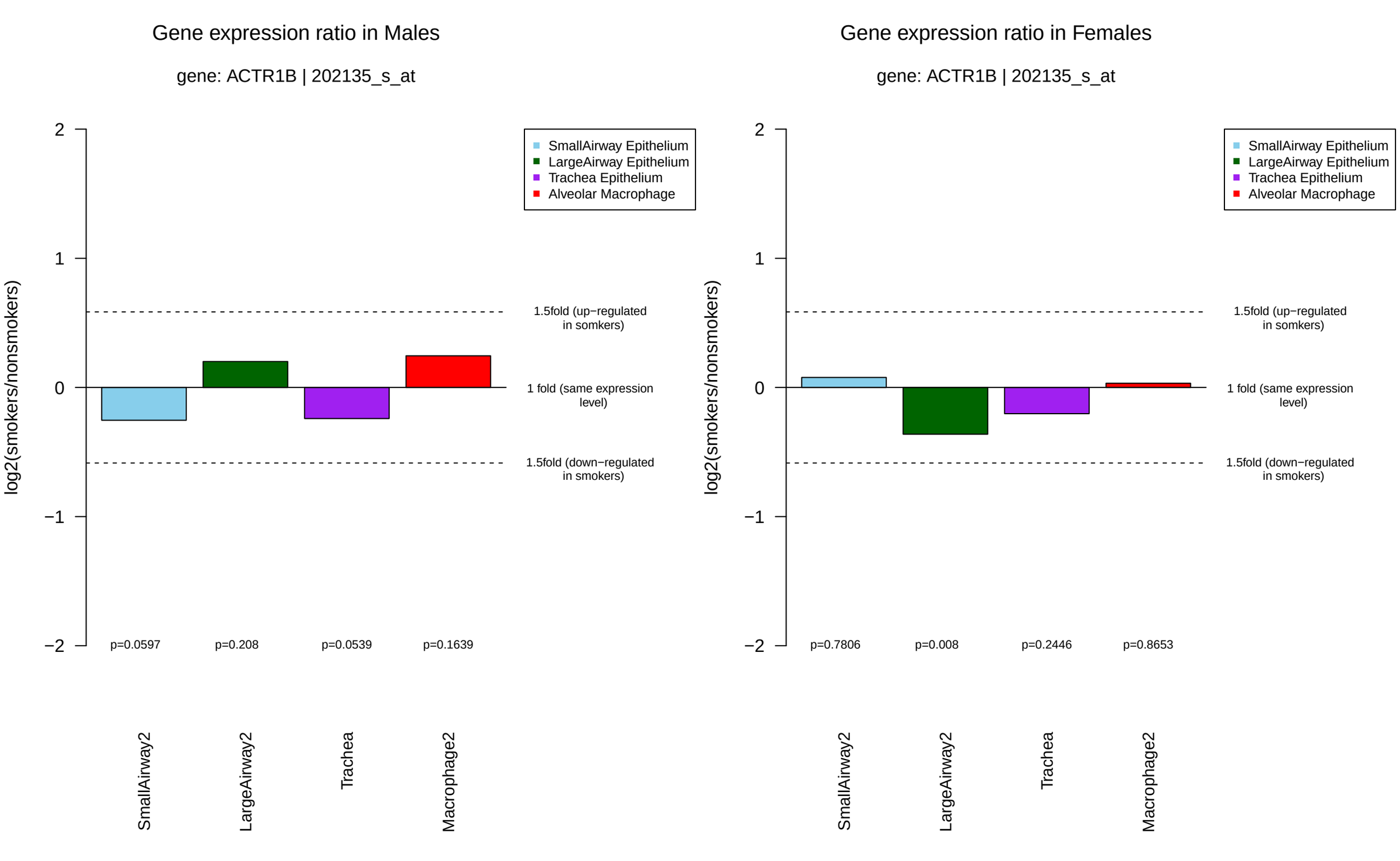
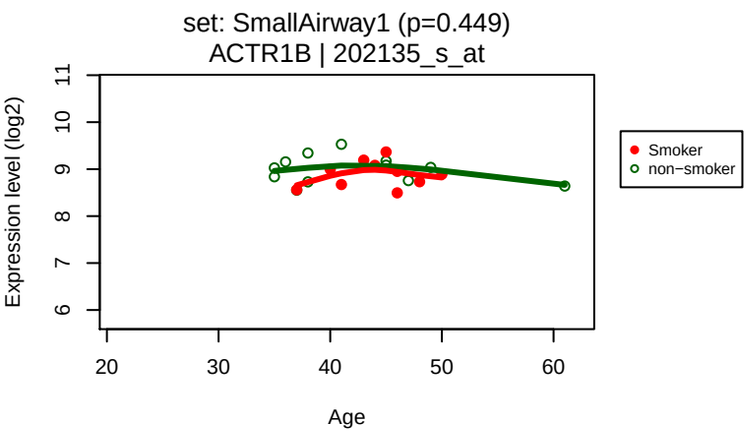
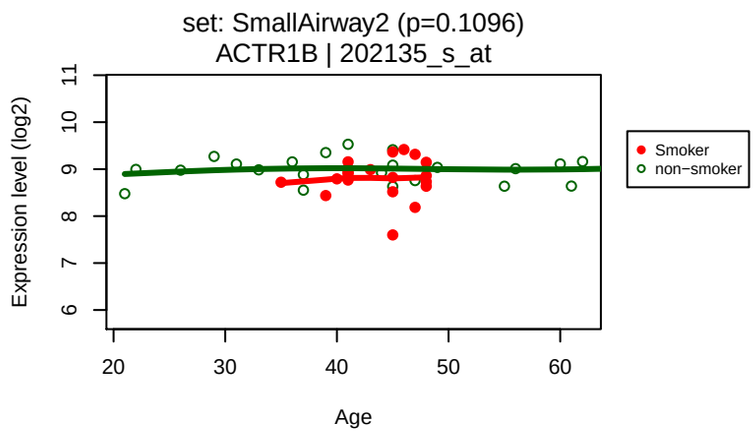
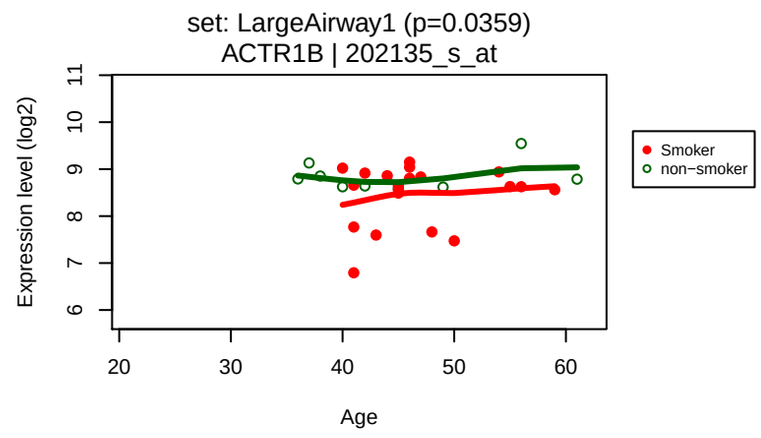

Gene expression of smokers and nonsmokers
gene: ACTR1B | 202135_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.449
p=0.1096
p=0.0744
p=0.0359
p=0.8819
p=0.0175
p=0.9737
p=0.1724
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

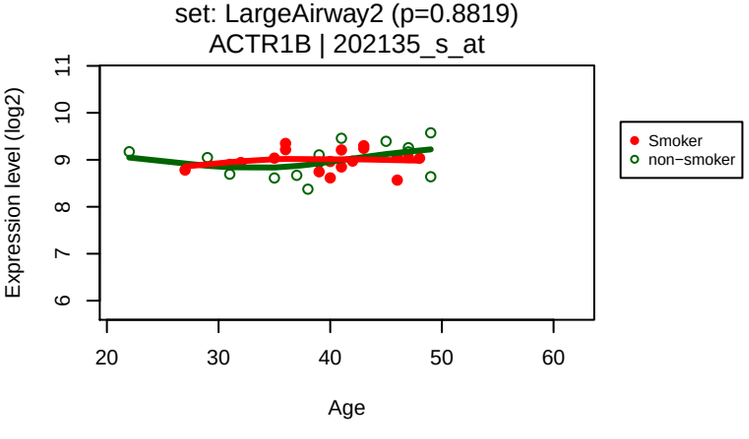

Gene expression ratio (smokers/nonsmokers)
gene: ACTR1B | 202135_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.449
p=0.1096
p=0.0744
p=0.0359
p=0.8819
p=0.0175
p=0.9737
p=0.1724
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

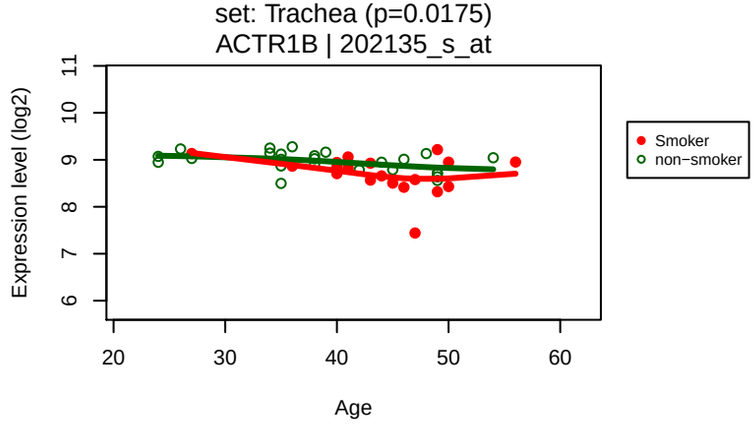

Gene expression ratio in Males
gene: ACTR1B | 202135_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0597
p=0.208
p=0.0539
p=0.1639
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTR1B | 202135_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7806
p=0.008
p=0.2446
p=0.8653
SmallAirway2
LargeAirway2
Trachea
Macrophage2

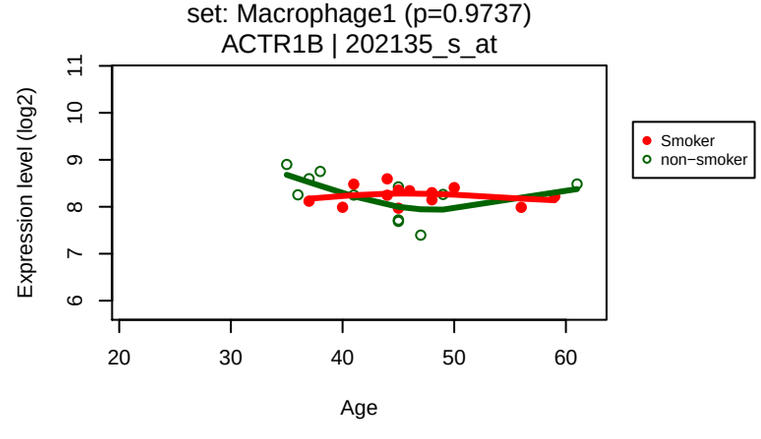

set: SmallAirway1 (p=0.449)
ACTR1B | 202135_s_at
Expression level (log2)
Age
Smoker
non−smoker

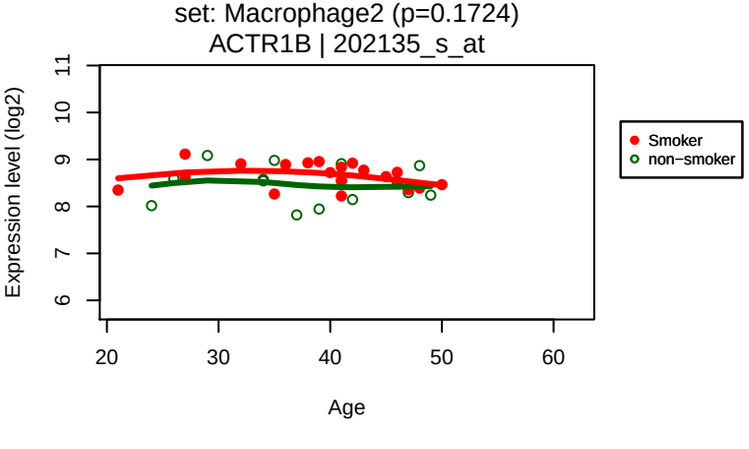

set: SmallAirway2 (p=0.1096)
ACTR1B | 202135_s_at
Expression level (log2)
Age
Smoker
non−smoker

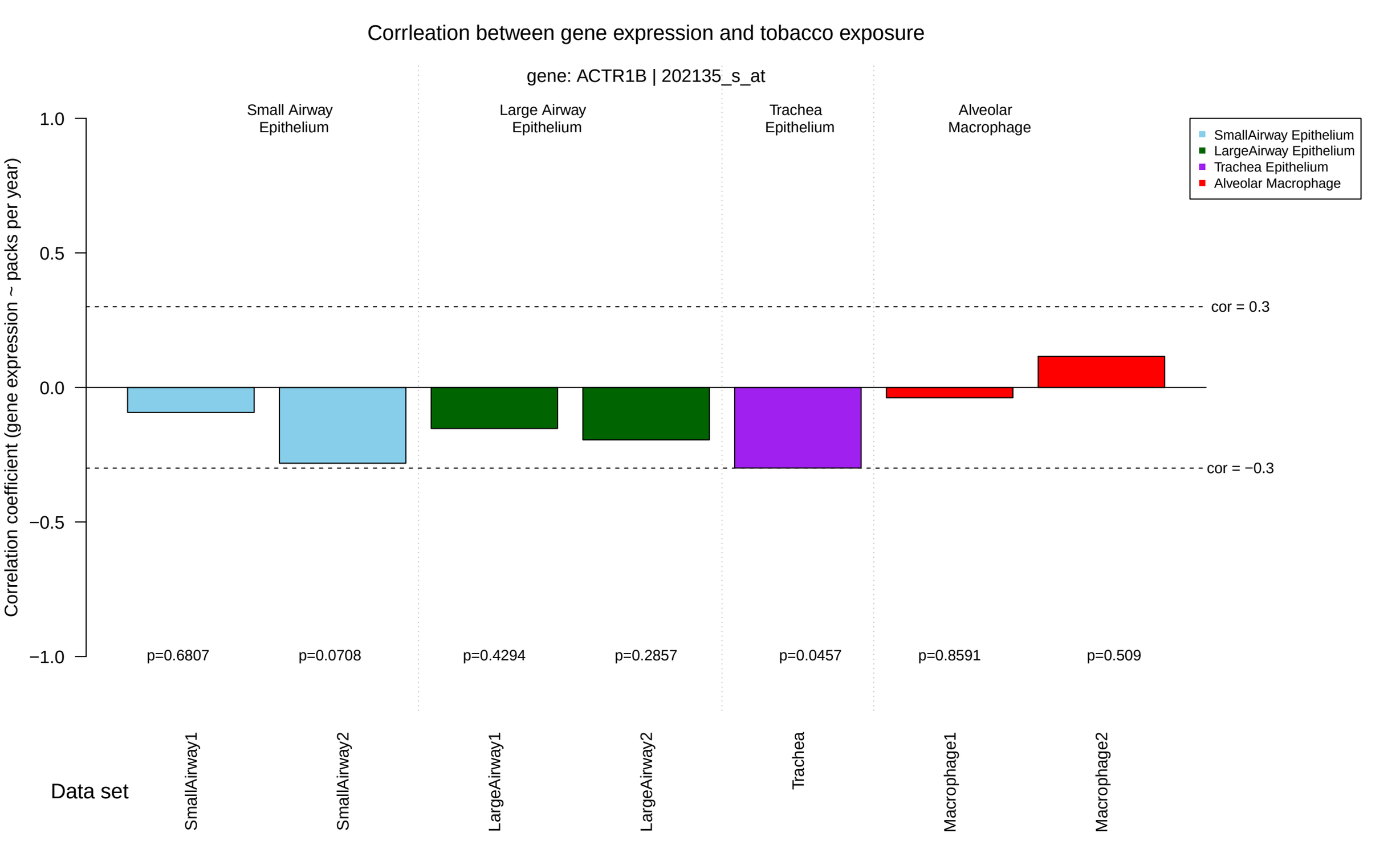

set: LargeAirway1 (p=0.0359)
ACTR1B | 202135_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8819)
ACTR1B | 202135_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0175)
ACTR1B | 202135_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9737)
ACTR1B | 202135_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1724)
ACTR1B | 202135_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTR1B | 202135_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6807
p=0.0708
p=0.4294
p=0.2857
p=0.0457
p=0.8591
p=0.509
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2